

Confidence Intervals

Grinnell College

October 9, 2026

Question

Suppose I am interested in measuring sticks down at Arbor lake, and I know the true average length of the sticks is 14 inches. If I plan on collecting 500 samples of size n , how could I find an interval that will contain the middle 80% of my sample means?

1. If we increase n , should this interval get wider or narrower? Why?
2. If individual stick lengths become more variable, what happens to the interval?
3. About how many of our 500 sample means would fall outside this interval? Would that mean something went wrong?

From this week we remember...

1. **Central Limit Theorem**
2. **Standardization**
3. **Standard Normal**

Benefits of a distribution

Empirical Distribution:

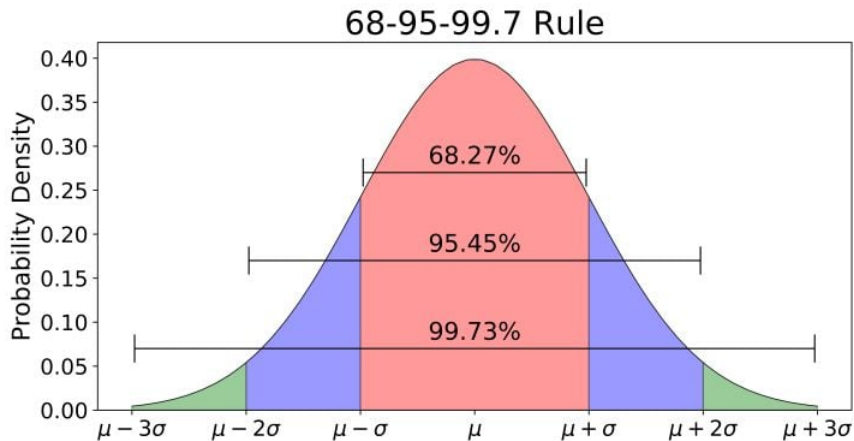
Suppose I have gone out and collected a sample:

- ▶ If I wanted to find the median of this dataset, what would I do?
- ▶ What if I wanted to find Q1 and Q3 of this dataset?

Named Distribution:

What if instead I wanted to find the median and Q1 and Q3 of a normal distribution with mean value μ and standard deviation σ ?

Empirical Rule



Determining Range

If we can determine that the probability of our standardized values being between -1 and 1 to be

$$P(-1 < Z < 1) = 0.68$$

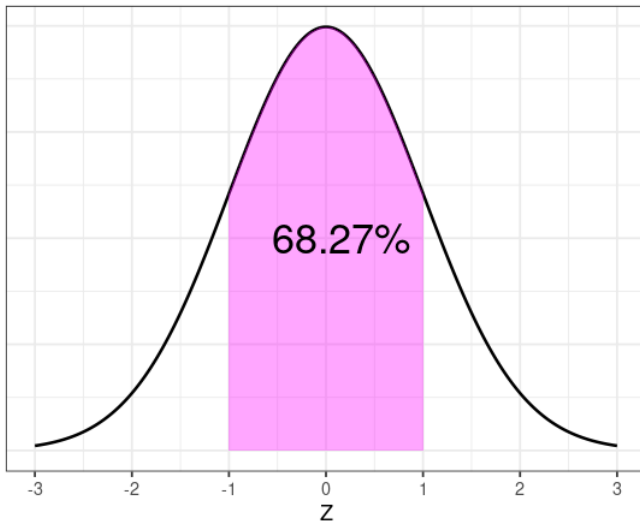
and

$$P(-2 < Z < 2) = 0.9545$$

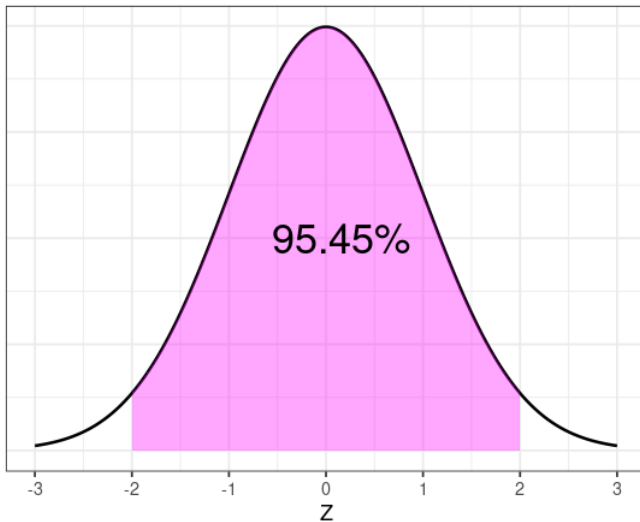
Then theoretically to get the middle $M\%$ of values we should be able to find values C such that

$$P(-C < Z < C) = M\%$$

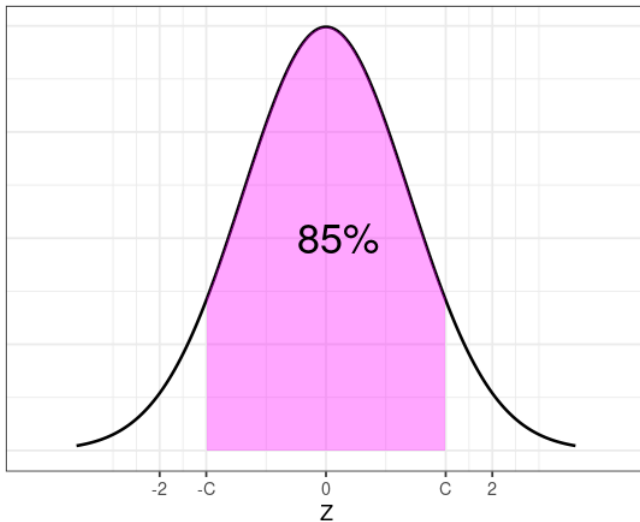
$$P(-1 < Z < 1) = 0.68$$



$$P(-2 < Z < 2) \approx 0.95$$



$$P(-C < Z < C) = 0.85$$



Critical Values

In general, we can create a symmetric interval around 0 such that a randomly collected standardized sample mean Z will fall within this interval with some probability

$$P(-C < Z < C) = M$$

The selected probability, M , is defined as our **confidence**; this is the frequency with which our statistic will fall within this interval

The bounds of this interval, C , is called our **critical value**. *The choice of C is the only thing we choose that determines our level of confidence*

But we don't know μ

Suppose we wanted to construct an interval that 68% of the time would contain the true parameter μ :

$$\begin{aligned}68\% &= P(-1 < Z < 1) \\&= P\left(-1 < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < 1\right) \\&= P\left(-\frac{\sigma}{\sqrt{n}} < \bar{X} - \mu < \frac{\sigma}{\sqrt{n}}\right) \\&= P\left(-\bar{X} - \frac{\sigma}{\sqrt{n}} < -\mu < -\bar{X} + \frac{\sigma}{\sqrt{n}}\right) \\&= P\left(\bar{X} - \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + \frac{\sigma}{\sqrt{n}}\right)\end{aligned}$$

In general,

$$\begin{aligned}M\% &= P(-C < Z < C) \\&= P\left(-C < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < C\right) \\&= P\left(\bar{X} - C \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + C \frac{\sigma}{\sqrt{n}}\right)\end{aligned}$$

This gives us a general method for constructing an $M\%$ confidence for μ of the form

$$\bar{X} \pm C \times \frac{\sigma}{\sqrt{n}}$$

But why?

There are several things we can do with this machinery that are both different and basically the same:

- ▶ Sometimes we know μ and σ , and we want to determine a range of where our sampled $\bar{X}$ will be
 - ▶ $\bar{X}$ this helps us determine if a particular sample mean is far from what we expect
- ▶ Sometimes, we only know $\bar{X}$ and need to find an interval for μ
 - ▶ This allows us to give an estimate for μ that uses the statistic $\bar{X}$ and information about our uncertainty

Both of these goals use the fact that $\bar{X}$ is normally distributed

Confidence versus Precision

We want an interval that contains the population mean reliably, but is also narrow enough to be useful.

$$\bar{X} \pm C \frac{\sigma}{\sqrt{n}}$$

- ▶ **Want more confidence?** Increase C . The interval gets wider.
- ▶ **Want a narrower interval at the same confidence?** Increase n . The standard error gets smaller.
- ▶ **Why standardize?** It lets us use the same normal cutoff C across different sample sizes and population standard deviations.

A few notes

Assuming we also know σ , we are able to construct a range of plausible values for μ based on information from our sample, namely $\bar{X}$ and n that is of the form:

$$\bar{X} \pm \text{Margin of error}$$

where this margin of error is determined by:

1. The sample size
2. The population standard deviation
3. The amount of confidence we wish to have in our interval

In other words, our margin of error is expressed by the term:

$$C \times \frac{\sigma}{\sqrt{n}}$$

Our basic goal is this: *can I use information from my sample to provide a range of reasonable estimates for my population parameter?*

In actuality, this represents two competing goals:

1. I want an interval that is narrow enough to provide a useful estimate of my parameter
2. I want an interval that is wide enough that it is likely to contain the true value

Confidence Intervals

Suppose the mean of my population is $\mu = 50$ with $\sigma = 15$. The statistics from a collected sample are

$$\bar{x} = 46.35, \quad n = 30$$

From here, we can construct a **95% confidence interval** of:

$$\begin{aligned} 95\%CI &= \text{Point estimate} \pm \text{Margin of Error} \\ &= \bar{x} \pm 2 \times \sigma / \sqrt{n} \\ &= 46.35 \pm 2 \times 2.74 \\ &= (40.87, 51.83) \end{aligned}$$

What does this even mean?

- ▶ 95% what?
- ▶ We are 95% sure it contains the mean?
- ▶ The probability of the mean being there is 95%?
- ▶ Or something else?

Repeated Samples

Suppose that I had repeated this process of collecting samples of size $n = 30$ and constructing intervals many times

$$\mu = 50, \quad \sigma/\sqrt{n} = 2.74$$

Sample	$\bar{x}$	95% CI	Contains μ
1	46.49	(41.01, 51.97)	TRUE
2	50.06	(44.58, 55.54)	TRUE
3	48.74	(43.26, 54.22)	TRUE
4	48.26	(42.78, 53.74)	TRUE
5	52.48	(47.00, 57.96)	TRUE
6	49.75	(44.27, 55.23)	TRUE
7	50.37	(44.89, 55.85)	TRUE
8	47.20	(41.72, 52.68)	TRUE
9	56.06	(50.58, 61.54)	FALSE
10	46.47	(40.99, 51.95)	TRUE

Confidence Intervals

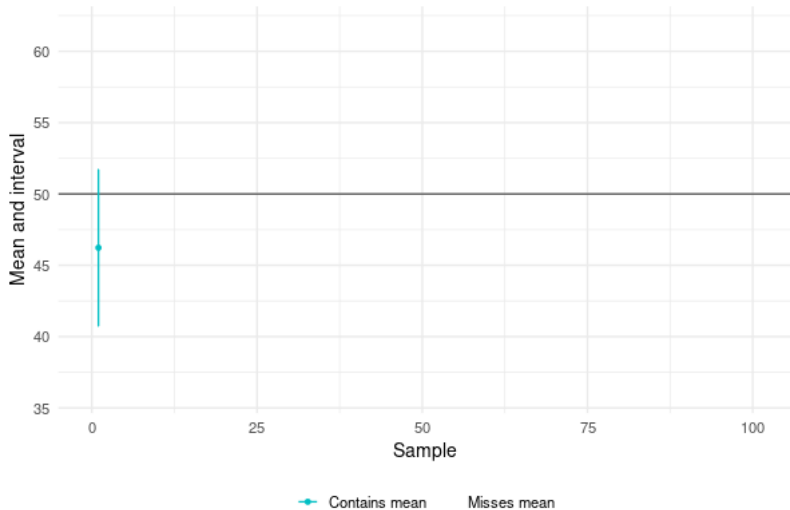
A **confidence interval** is an interval that has the following properties:

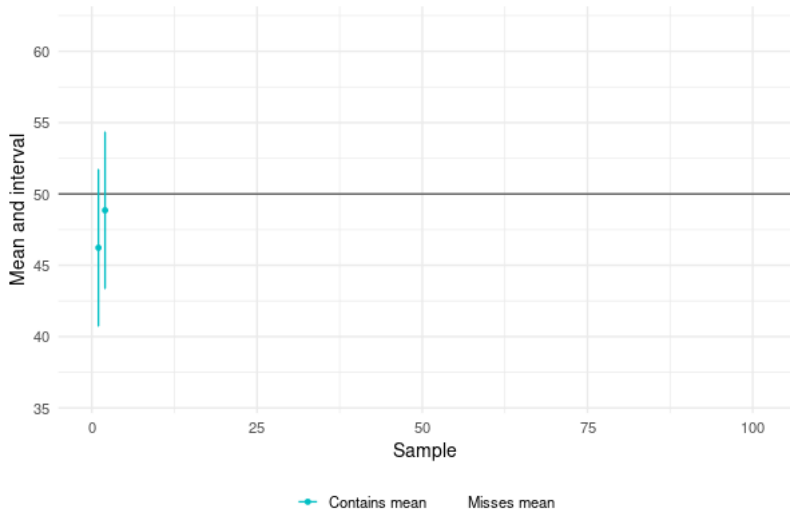
- ▶ It is the result of a *random process*
- ▶ It is constructed according to a procedure or set of rules
- ▶ It is made with the intention of giving a plausible range of values for a *parameter* based on a *statistic*

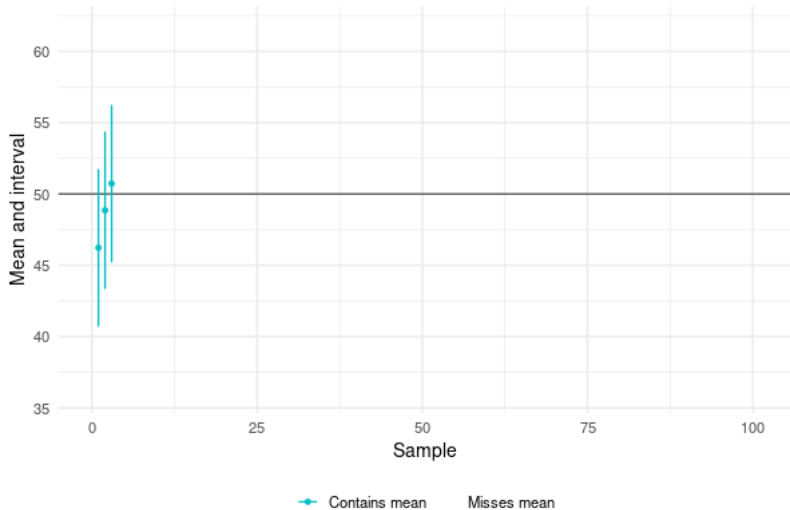
95% Confidence Interval

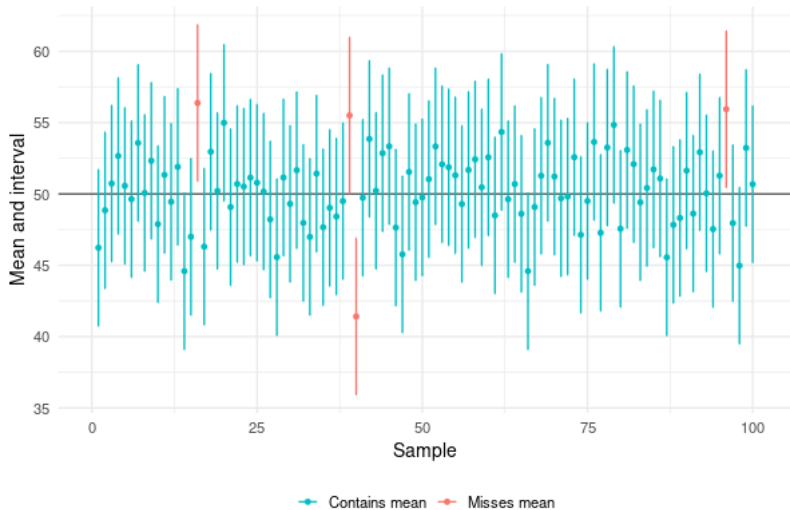
When we say something has a 95% confidence interval, what we mean is:

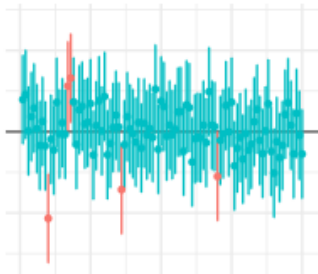
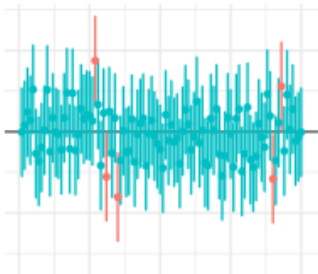
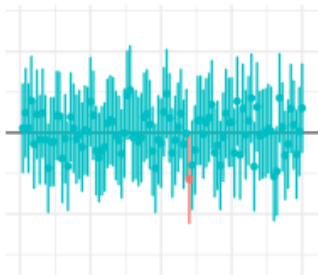
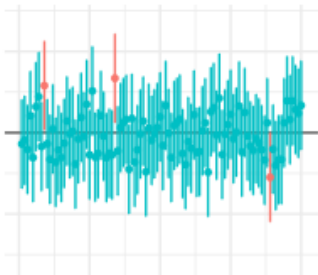
The process that constructed this interval has the property that, on average, it contains the true value of the parameter 95 times out of 100











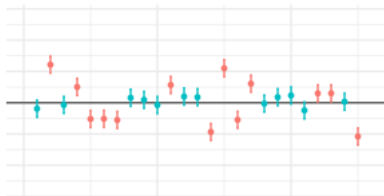
Confidence Intervals

It is also worth observing that we can *alter* our process to achieve different results. There is a trade-off between how frequently we are correct and how much uncertainty we allow in our prediction

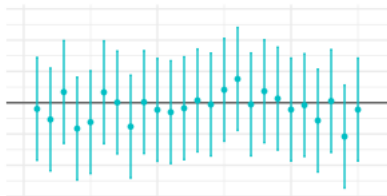
Mean ± 0.5 SE



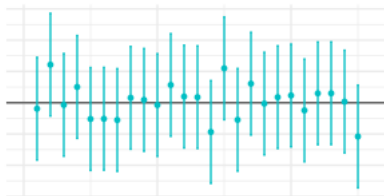
Mean ± 0.5 SE



Mean ± 3 SE



Mean ± 3 SE



Confidence Intervals

To be absolutely clear: we will generally *not* know if the confidence interval we construct contains the true value of the parameter

This is akin to throwing a dart but never seeing the target

This is the nature of statistical inference: we can describe properties of the *process* that created our intervals, but we can never conclusively speak about the interval itself

- ▶ **Standard deviation** (σ) is the standard deviation of our population, while **standard error** ($\sigma/\sqrt{n}$) is an estimate of the variability in estimating a parameter
- ▶ A **sampling distribution** describes the distribution of a statistic or parameter estimate if we could repeat the sampling process as many times as we wish
- ▶ Approximations to the normal distribution generally follow the **68-95-99.7 rule** with 1/2/3 standard deviations of the mean
- ▶ If these properties hold, we can create a reasonable interval of possible parameter values of the form Point Estimate $\pm$ Margin of Error
- ▶ A **confidence interval** is an interval with the properties that:
 - ▶ It is constructed according to a procedure or set of rules
 - ▶ It is intended to give plausible range of values for a *parameter* based on a *statistic*
 - ▶ The confidence level describes the long-run probability that intervals will cover the true parameter